

ZHIHAN GUO

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RESEARCH INTERESTS

I mainly worked on database system, with a focus on transaction processing and distributed databases, and I also have potential interest in cloud-native database and serverless computing.

EDUCATION

Ph.D. in Computer Science, UW-Madison / 2018.09 – 2023.05 (expected)

advised by Professor Xiangyao Yu, GPA: 3.95/4.00

B.S. (Hons.) in Computer Science, UW-Madison / 2015.09 – 2018.05

GPA: Overall 3.94/4.00; Major 3.96/4.00

AWARDS AND ACKNOWLEDGEMENTS

Microsoft Ph.D. Fellowship (US & Canada) / 2021

Dean's List (repeated), UW-Madison / Fall 2015 – Spring 2018

UW-Madison Undergraduate Scholarship for Summer Study, UW-Madison / Summer 2016

First-class Scholarship for Excellent Academic Performance (2013-2014), BLCU / Fall 2014

Excellent Leadership Award, BLCU / Spring 2014

PUBLICATIONS

Atomic Commit for Cloud DBMS with Storage Disaggregation

First Author

(under submission, title altered)

How Good is My HTAP System?

Non-First Author

(under submission, title altered)

Releasing Locks as Early as You Can: Reducing Contention of Hotspots by Violating Two-Phase Locking

Zhihan Guo, Kan Wu, Cong Yan, Xiangyao Yu,

SIGMOD 2021

The Storage Hierarchy is Not a Hierarchy: Optimizing Caching on Modern Storage Devices with Orthus

Kan Wu, **Zhihan Guo**, Guanzhou Hu, Kaiwei Tu, Ramnatthan Alagappan, Rathijit Sen, Kwanghyun Park, Andrea Arpaci-Dusseau, Remzi Arpaci-Dusseau,

FAST 2021

The Storage Hierarchy is Not a Hierarchy: Optimizing Caching on Modern Storage Devices with Orthus

Kan Wu, **Zhihan Guo**, Guanzhou Hu, Kaiwei Tu, Ramnatthan Alagappan, Rathijit Sen, Kwanghyun Park, Andrea Arpaci-Dusseau, Remzi Arpaci-Dusseau,

NVMW 2021

A Statistical Perspective on Discovering Functional Dependencies in Noisy Data
Yunjia Zhang, **Zhihan Guo**, Theodoros Rekatsinas,
SIGMOD 2020

Unsupervised Functional Dependency Discovery for Data Preparation
Zhihan Guo, Theodoros Rekatsinas,
ICLR, ILD Workshop 2019

WORKING EXPERIENCE

Affiliate Research Assistant at Microsoft Gray System Lab, Madison / Fall 2021

Software Engineer Intern at Facebook, Delos Team, Madison / Summer 2021

supervised by Dr. David Geraghty, directed by Dr. Mahesh Balakrishnan

RESEARCH EXPERIENCE

UW Madison Database Group / 2018.02-Present

A Survey on Hotspot Techniques in Concurrency Control [ongoing]

advised by Prof. Xiangyao Yu, Dr. Philip Bernstein (@MSR)

Examined the techniques in concurrency control that address the issues of hotspots and developing microbenchmarks to evaluate key characteristics of the techniques.

Improve Two-Phase Commit for Databases with Storage Disaggregation

advised by Prof. Xiangyao Yu, Dr. Philip Bernstein (@MSR)

Addressed the blocking issue and reduced one logging in two-phase commit by leveraging the key features provided by storage disaggregation architecture.

Improving Two-Phase Locking to Reduce Contention with Hotspots

advised by Prof. Xiangyao Yu

Proposed a new concurrency control protocol and three optimization techniques as an extension to 2PL to increase concurrency when hotspots are present.

Learning Functional Dependencies over Noisy Data via Sparse Regression

advised by Prof. Theodoros Rekatsinas

Proposed a FD discovery approach over noisy data by casting the problem as structure learning over probabilistic graphical model and solving it through sparse regression.

Facebook / 2021.05 - 2021.11

Characterizing Storage for Control Plane Services at Facebook [ongoing]

supervised by Dr. David Geraghty, advised by Dr. Mahesh Balakrishnan

Characterized the workload in various use cases; Explored the design implications for Delos based on the analysis across three layers in Delos – key-value interface, local storage (RocksDB), the shared log layer (state machine replication).

Wisconsin Human-Computer Interaction Laboratory / 2017.02 – 2018.05

Robots Providing Emotional Support (Senior Honors Thesis)

advised by Prof. Bilge Mutlu

Used a programmable robot to study if non-humanoid robot with non-language vocalization can act as a listener in social sharing process to reduce negative effects.

Child Emotion Lab / 2016.01 – 2017.09

Facial Cues for Emotion Recognition

supervised by Brian Letizke, directed by Prof. Seth Pollak

Communicated with parents and children from local community; Conducted psychological experiments and collected physiological data.

TEACHING EXPERIENCE

Teaching Assistant for Database Management Systems, UW-Madison / Fall 2019

supervised by Prof. Goetz Graefe

SKILLS & RELEVANT COURSEWORK

Programming language: C++, C, Python, Java, MATLAB.

Relevant coursework: database, operating system, distributed system, machine learning